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The effect of mechanical and thermal properties of FRP bars on their tensile performance under elevated temperatures

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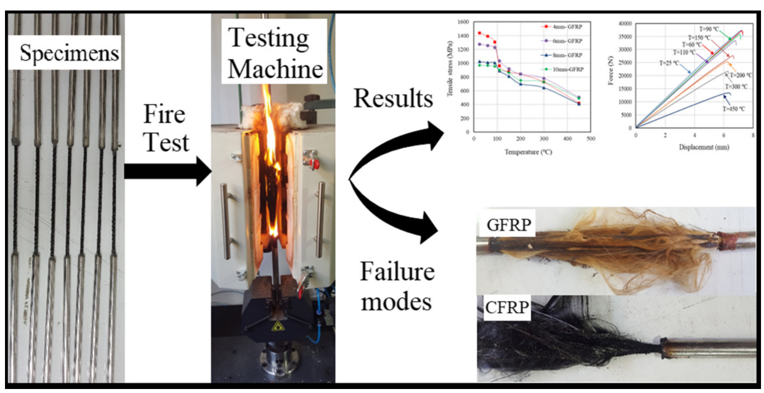
HIGHLIGHTS

- Effects of physical and thermal properties on the performance of FRP bars under elevated temperatures were studied.
- Temperatures below the FRP bars' T_g slightly affect their tensile behavior.
- At very high temperatures the ultimate tensile strengths decreased considerably.
- The bars with a larger diameter showed better tensile performance at elevated temperatures.

GRAPHICAL ABSTRACT



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ABSTRACT

In this experimental study the effect of physical and thermal properties of various FRP bars on their performance under elevated temperatures are investigated. The parameters included the bars' diameter, type of fiber, type of resin, fiber to matrix ratio, and thermal properties were studied. Moreover, ANOVA (ANalysis Of VAriance) was performed in order to investigate the contribution of each variable on the obtained results. The results showed that in addition to the temperature, the bars' diameter, type of fiber, type of resin, and thermal properties (T_g and T_d) of the FRP bars have contributions to the results, while the fiber to matrix ratio was found to be an ineffective factor. It was also found that the FRP bars experienced significant reductions in their ultimate tensile strength upon exposure to extreme temperatures of up to 450 °C, at which they lost about 50–70% of their reference tensile strength.

1. Introduction

The corrosion of traditional steel reinforcing bars is known as the main factor responsible for premature reinforced concrete deterioration and loss of service life. It is a result of exposure to harsh environments, such as sea water, industrial chemicals, and deicing salts [1–3]. One promising solution to this problem is using fiber-reinforced polymer (FRP) reinforcing bar as a corrosion- resistant material instead of steel bar [4–6]. Aside from being corrosion-resistant, other outstanding characteristics, such as high stiffness and high strength-to-weight ratio, appropriate fatigue performance, electric insulation, and ease in fabrication, have made them ideal materials used to eliminate durability problems

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associated with steel-reinforced structures in aggressive environments [7–10]. Therefore, recently, FRP bars have been successfully used to reinforce concrete members in the construction of roads and bridges [11,12]. On the other hand, relatively material cost, low ductility, and consequent brittle failure, poor mechanical properties of the resin, low shear strength, and high vulnerability of strength and stiffness at elevated temperatures are some of the FRP material's disadvantages that must be considered when using them in construction applications [13,14]. In order to achieve wider acceptance and application of such materials in the construction industry, the possibility of exposure to fire of such concrete structures is an issue that must be looked into seriously during the design of a structure [15,16]. However, data regarding the fire resistance of FRP-reinforced concrete structures (e.g., how long the structure can maintain its overall integrity at high temperatures, as well as fire exposures, and the amount of bond, strength, and stiffness decrease at elevated temperatures) are very limited.

Based on the available literature, Hamad et al. [17] conducted a study on the effect of elevated temperatures on the mechanical properties of different FRP bars and reported about 55% and 30% loss in their tensile strength and elastic modulus, respectively, at a critical temperature of 325 °C. Further, Wang et al. [18] showed tensile strength reductions of about 45% and 35% of glass fiber-reinforced polymer (GFRP) and carbon fiber-reinforced polymer (CFRP) bars, respectively, at 350 °C, compared to their original ambient temperature strengths. The effects of elevated temperatures on the properties of GFRP bars were also studied by Robert and Benmokrane [15], they reported a reduction of 40% in tensile strength at a temperature of 250 °C. Under fire conditions, when FRP materials first reach their glass transition temperature, T_g (about 65°–120 °C), their resin changes from a glassy state to a rubbery state. After that, at higher temperatures, when FRP composites reach their resin decomposition temperature, T_d (about 300°–400 °C), their chemical bonds, modular chains of the resin, and bonds between fibers may break. Finally, at higher temperatures, resin ignition occurs [13,17–20]. However, the fibers that have better thermal properties compared to the resin can still carry some load in the longitudinal direction until they reach their threshold temperature (e.g., about 1000 °C for glass fibers) [15]. Due to a lack of experimental data on the thermal mechanical properties of a wide variety of FRP composites and their mechanical degradation under fire conditions, the critical temperature of FRP materials that affects their properties is not accurately defined [17,21]. According to the Canadian standards, CAN/CSA-S806-02 [22], when an FRP-RC loses 50% of its initial strength, the structure fails. Therefore, the critical temperature at which an FRP bar loses about half of its tensile strength should be obtained in order to have a more accurate design of FRP-RC member [17].

With these in mind, since scant experimental data are available on the degradation of the mechanical properties of FRP bars under fire conditions, this study investigates the effects of fiber type (GFRP and CFRP), resin type (vinylester and epoxy), fiber-to-resin volumetric ratio, and bar diameter on the fire resistance of the FRP bars used as internal reinforcement for concrete. Moreover, the critical temperature of each bar was investigated. The findings of this study will contribute to better understanding the effects of different parameters on the fire resistance of FRP bars subjected to fire conditions, and thus to modify the current FRP standards and guidelines for their reduction factors.

2. Experimental program

A series of specimens was tested in tension in order to investigate the effects of different parameters-namely fiber type, resin type, and bar diameter on the fire resistance behavior (the critical temperature and mechanical properties) of FRP bars used as internal reinforcement for concrete.

2.1. Materials

2.1.1. FRP bars

Four types of FRP reinforcement bars were used in this study, including sand-coated GFRP reinforcing bars (Type A), helically wrapped CFRP reinforcing bars (Type B), and grooved CFRP bars with two different resins (Type C and Type D), as shown in Fig. 1. Sand-coated GFRP bars, had nominal diameters of 4, 6, 8, and 10 mm, while all CFRP bars had a nominal diameter of 5 mm. The thermal and mechanical properties of the FRP bars, as provided by their manufacturers, are listed in Table 1.

2.1.2. High strength adhesive

A high-strength adhesive was used to adhere grip circular steel tubes to the FRP bars' free ends for conducting tensile tests. These steel pipes, which provide an adequate confinement pressure on the bar, were used in order to prevent bar slippage or local failure due to the low gripping resistance of FRP bars. The shear strengths of this adhesive, as provided by the manufacturer, was 36 MPa.

2.2. Specimens

GFRP bar specimens of 800 mm length and 4, 6, 8, and 10 mm diameter and CFRP bar specimens of 800 mm length and 5 mm diameter were prepared for tensile tests. A total of 160 specimens was constructed. Three identical specimens were prepared for each condition in order to obtain reliable results. The ends of the specimens were anchored using steel pipes filled with high strength adhesive. The minimum lengths and diameters of the pipes were determined according to the bar's ultimate tensile strength (at ambient temperature) and diameter, respectively. The steel pipes' length, outer diameter, and thickness are shown in Fig. 2. In order to make sure that the bars are embedded at the center of the pipes, the bars were kept at the center using an aluminum cap at both pipe's end.

2.3. Conditioning of specimens

The FRP bars were separated into unconditioned series, used as controls, tested at ambient temperature, and conditioned series, tested at elevated temperatures. Based on the previous studies [15,17], temperatures of 25°, 60°, 90°, 110°, 150°, 200°, 300°, and 450 °C were chosen to simulate short-term condition in the case of fire. According to the study of Kodur and Bisby [23] on the fire resistance of FRP-reinforced concrete slabs, approximately 30, 60, and 185 min for concrete depths of 15, 30 and 75 mm, were predicted as the time for FRP bars to reach 350 °C. Therefore, a steady state regime was used in this study.

2.4. Tensile test methodology

The tensile tests were conducted using a UTM Santam servo electric testing machine with a three-zone split electric furnace according to ASTM standards [24] recommendations (Fig. 3). Displacement controlled loading at a rate of 1.2 mm/min was applied to the specimens. In order to determine the exact elastic modulus of the FRP bars, an extensometer was attached to the middle of the control bars. The bars were heated up at a heating rate of 10 °C/min until reaching the

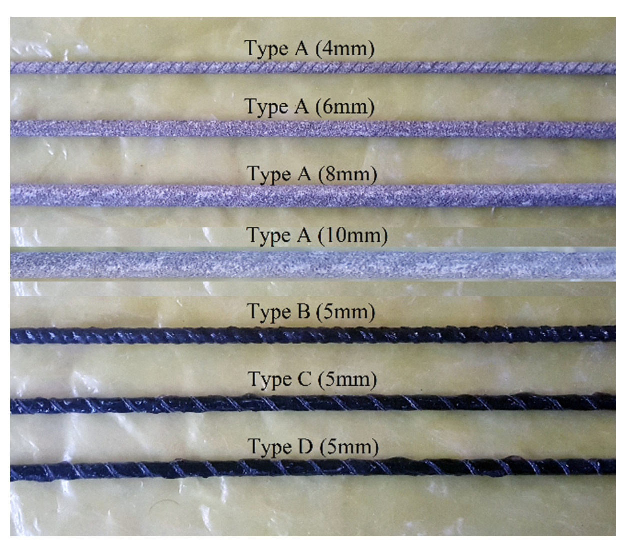

Fig. 1. FRP bars used in this study.

Table 1
Material properties of FRP bars used in this study.

Property	Bar type			
	A	B	C	D
Resin type	Epoxy	Epoxy	Epoxy	Vinylester
Fiber volume (%)	75	75	85	85
Characteristic tensile strength (MPa)	1200	2100	1900	1700
Modulus of elasticity (GPa)	50	150	125	117
Strain at failure (%)	2.4	1.4	1.5	1.4
Density (g/cm^3)	2.20	1.65	1.70	1.70
Longitudinal coefficient of thermal expansion (1/°C)	9×10^{-6}	-8×10^{-6}	-8×10^{-6}	-8×10^{-6}
Transverse coefficient of thermal expansion (1/°C)	22×10^{-6}	85×10^{-6}	85×10^{-6}	85×10^{-6}
Longitudinal thermal conductivity (W/m°C)	3.12	70	70	70
Transverse thermal conductivity (W/m°C)	0.25	0.20	0.20	0.20
Glass transition temperature T_g (°C)	110	110	110	105
Cure ratio (%)	97.5	98	98	98

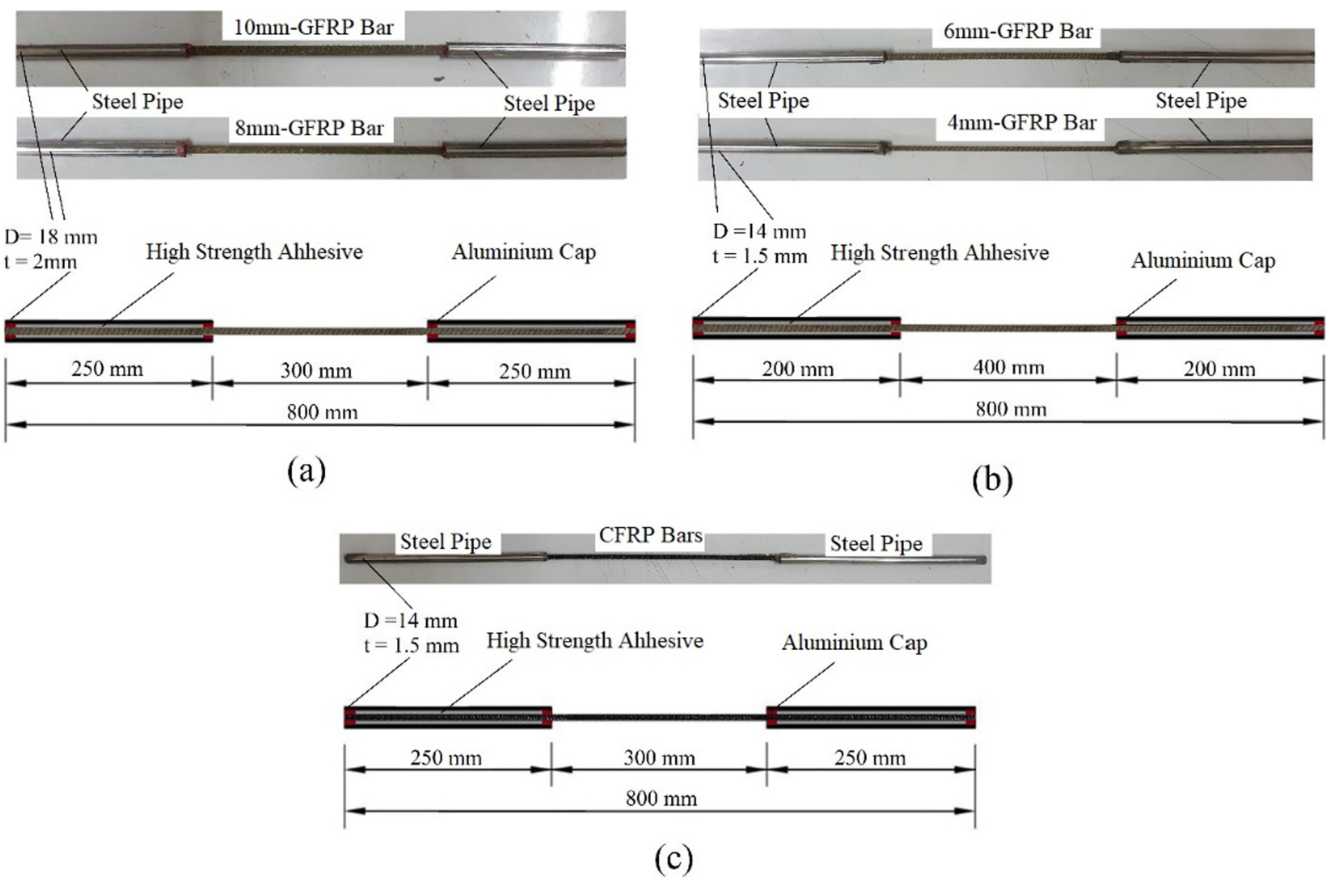

Fig. 2. Specimens' configuration: a) 8 mm and 10 mm-GFRP bars, b) 4 mm and 6 mm-GFRP bars, and c) all CFRP bars.

desired temperatures. Before starting the tests, similar to the study of Hamad et al. [17], the specimens were maintained in that target temperature for 30 min in order to obtain a uniform temperature distribution in the bars.

3. Test results and discussion

In this section, the results are presented and analyzed in detail with regard to the tensile strengths of the FRP bars. Regarding elevated temperatures, it is known that when the temperature reaches the glass transition temperature (T_g) of the matrix and above it, the resin softens, and thus the force transfer capacity between the fibers and resin reduces [25]. Generally, in a composite material, the fibers show better thermal properties compared to the matrix, and thus may continue to carry some loads in the longitudinal direction at very high temperatures. However, fiber/matrix debonding which decreases the load transferring between the fibers and resin may lessen the tensile strength of the overall composite with increased temperatures [26].

3.1. GFRP bars (Type A)

Table 2 presents the detailed tensile test results of the control and conditioned specimens of GFRP bars (Type A). It shows that, generally, the tensile strength of all types of FRP bars decreased as temperature increased. The typical tensile load versus displacement curves of the selected GFRP bars tested at different temperatures are depicted in Fig. 4. It should be noted that because the extensometer could get damaged during tests at elevated temperatures, the bars' strain and modulus elasticity were not measured

Fig. 3. Test setup for elevated temperatures.

for conditioned specimens. As Fig. 4 shows, all specimens showed a linear deformation until they approached their ultimate load, where the bar experienced a very slight non-linear deformation. Once the bar's load reached its maximum tensile capacity, a sudden drop, followed by complete failure, occurred in its curve.

The relation between the ultimate tensile stress of GFRP bars and the test temperatures, as well as that between tensile stress retention and temperature, are shown in Fig. 5. In Fig. 5b, the retention-temperature curves of GFRP bars can be separated into four different temperature zones. The first ranged from ambient temperature to 90 °C, which are temperatures below the T_g of the GFRP bars (T_g = 110 °C). In this zone, the molecular chain mobility of the matrix did not change, and thus, the ultimate tensile strength remained nearly stable [15]. This demonstrates that temperatures below the T_g do not significantly affect the tensile mechanical properties of such bars. The second zone included temperatures

Table 2
Tensile test results of GFRP bars at elevated temperatures.

Temperature (°C)	4 mm-GFRP			6 mm-GFRP			8 mm-GFRP			10 mm-GFRP		
	Average ultimate stress (MPa)	COV (%)	Retention (%)	Average ultimate stress (MPa)	COV (%)	Retention (%)	Average ultimate stress (MPa)	COV (%)	Retention (%)	Average ultimate stress (MPa)	COV (%)	Retention (%)
25	1438	1.5	100	1274	2.4	100	1024	3.4	100	971	1	100
60	1391	1.4	97	1256	2.1	99	1010	3.2	99	964	1.5	99
90	1304	4.1	91	1227	2.3	96	1005	4.2	98	955	0.6	98
110	960	2.9	67	1034	2.8	81	890	4.8	87	903	2.6	93
150	871	4.8	61	918	2.5	72	811	4.1	79	862	2.8	89
200	842	2.3	59	841	4.4	66	700	3.6	68	750	2.2	77
300	725	4.6	50	780	5.3	61	645	3.7	63	727	4.1	75
450	420	14.3	29	503	9.5	39	410	12.4	40	490	6.1	50

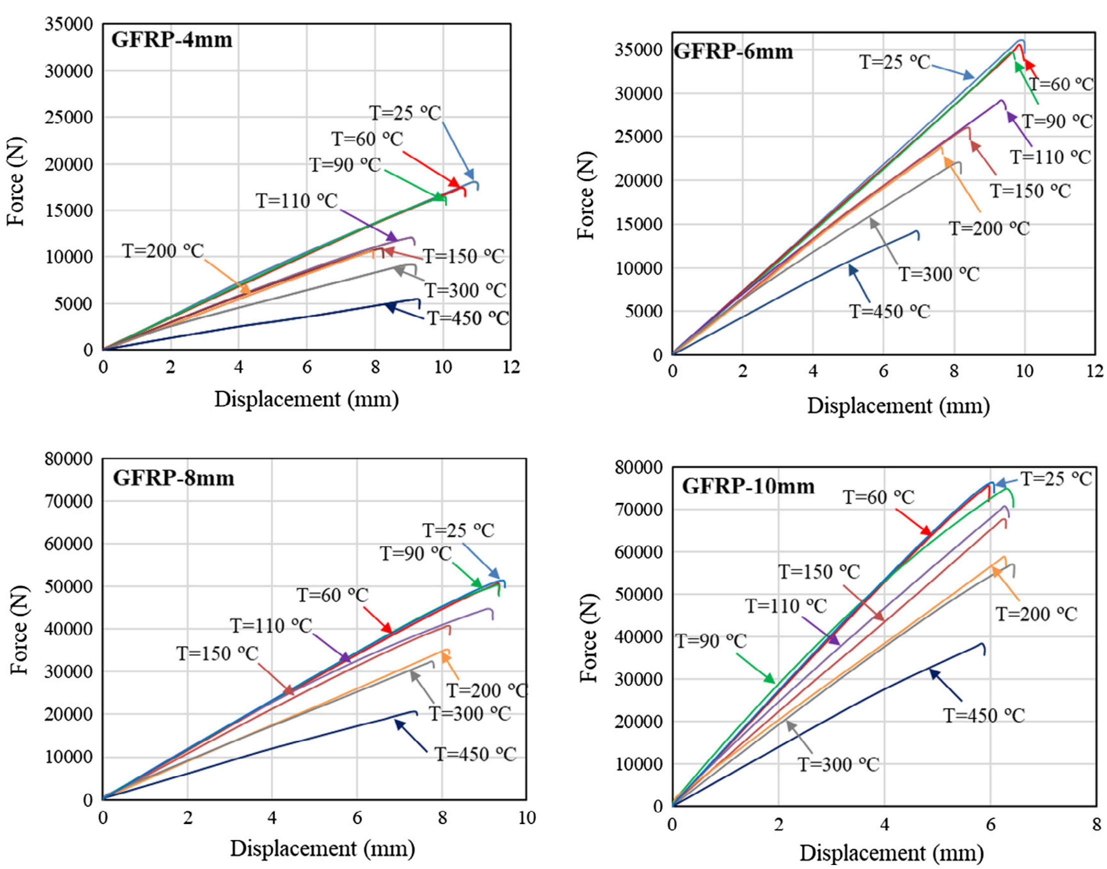

Fig. 4. Tensile load versus displacement curves of selected GFRP bars tested at different temperatures.

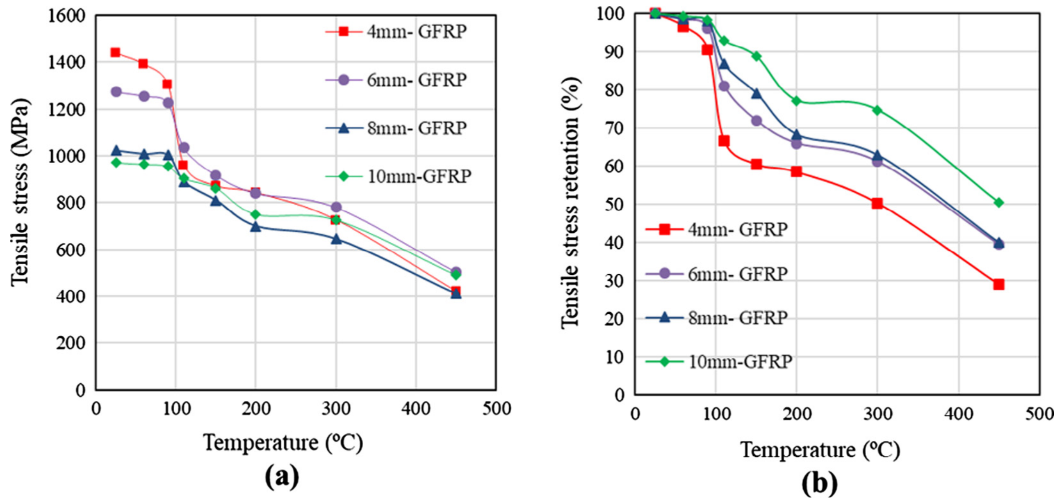

Fig. 5. GFRP bars' test results: a) Ultimate tensile stress vs the test temperatures, and b) Ultimate tensile stress retention vs test temperatures.

between 90 °C and 150 °C, which the temperatures approach T_g. At this zone, the resin softens, the molecular bonds break, and thus the force transfer capacity between the fibers and resin reduces. As a result, the ultimate tensile strength decreased considerably [17]. The third zone included the temperatures between 150 °C and 300 °C, which are below the bars' decomposition temperature (T_d). In this zone, the ultimate tensile strength experienced a slighter decrease in tensile strength compared to the second zone because the fibers continued to carry some loads in the longitudinal direction (the resin and fiber/matrix interface were damaged significantly), and at those temperatures, the damage of the fibers was not significant. The last zone is the temperatures higher than 300 °C, where temperatures were near and above the decomposition temperature of the bars. In this zone, the load carrying capacity of the fibers reduced more severely compared to the third zone.

A similar trend of tensile strength losses has been reported by other researchers [15,18,27]. Robert and Benmokrane [15] obtained a retention-temperature curve similar to the present study for GFRP bars, in which reductions of about 25%, 29%, 32%, and 46% of GFRP tensile strength when tested at 100°, 150°, 200°, and 300 °C, respectively, were reported. Wang et al. [18] also reported about 24%, 42%, 47%, and 65% decreases in temperature at 100°, 200°, 300 °C, and 400 °C, respectively, for 12.7 mm-GFRP bars. The trend of bar strength reduction was approximately the same as the present study (Fig. 5). Wang and Zha [27] also observed a similar behavior in GFRP bar reduction under elevated temperatures.

From the Fig. 5b, the critical temperatures of GFRP bars (the temperature in which the bar loses 50% of its tensile strength) were obtained approximately about, 300 °C, 375 °C, 377 °C, and 450 °C, for 4, 6, 8, and 10 mm bars, respectively. Therefore, it can be concluded that the larger bar diameter, the higher critical temperature. It is worth mentioning that the critical temperatures are representative values for specific GFRP bars used in this study, and they can be varied for other products. For example Kodur and Bisby [23] used a critical temperature of 325 °C for 12.7 mm GFRP bars in modeling FRP-RC structures.

Fig. 5 also shows that the bars with a larger diameter showed better tensile performance at elevated temperatures than those with a smaller diameter. This seems to be due to the lower fiber/-matrix detachment of the larger bars' core as a result of a higher confinement force applied by the outer bulk of the bar on its core and/or due to a lower combustion and oxidation of the inner core of larger bars (the outer layers prevent oxygen penetration into inner layers). Consequently, it seems that as a result of lower chemical reactions and higher confinement of the inner core of larger bars, the bars with a larger diameter are more resistant than smaller ones under elevated temperatures. Therefore, during the study of the mechanical properties of FRP bars under elevated temperatures and, consequently, their thermal stability, considering the effect of a bar's diameter is recommended.

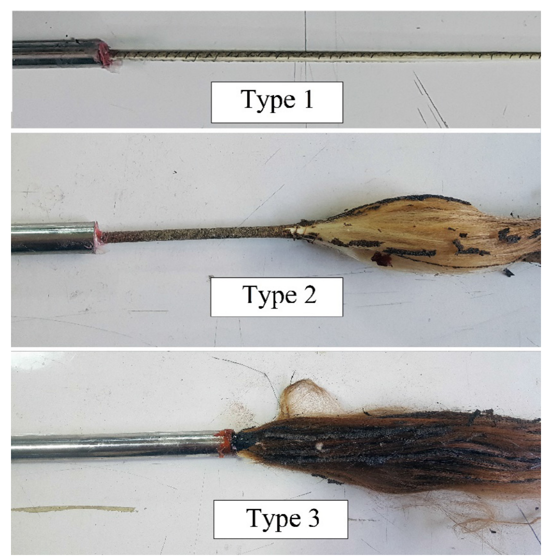

Fig. 6. Failure modes of GFRP bars after tensile tests under elevated temperatures.

Due to subjecting the bars to a combination of applied transverse compression and longitudinal extension [28], all specimens were splinted along their free length. Based on the observations (Fig. 6), the GFRP bars' failure modes can be classified into three types, based on the test temperatures. The first failure type was observed in the specimens tested at temperatures below 150 °C (zones 1 and 2, as mentioned above). The specimens tested at temperatures between 150 °C and 300 °C experienced the second failure type, in which the separation of fibers and resin is clearly visible (zone 3). The last failure type occurred in the specimens tested at temperatures exceeding 300 °C, whereby the bars became brown in color due to combustion and oxidation.

3.2. CFRP bars (Type B–D)

Detailed tensile test results pertaining to the control and conditioned CFRP bar specimens (Type B–D) are presented in Table 3.

Table 3
Tensile test results of CFRP bars at elevated temperatures.

Temperature (°C)	Type B			Type C			Type D		
	Average ultimate stress (MPa)	COV (%)	Retention (%)	Average ultimate stress (MPa)	COV (%)	Retention (%)	Average ultimate stress (MPa)	COV (%)	Retention (%)
25	2138	0.7	100	1905	0.9	100	1632	0.9	100
60	2110	1.3	98	1895	1.5	99	1621	1.0	99
90	1987	1.3	93	1824	2.2	96	1638	1.5	100
110	1807	2.7	84	1696	1.6	89	1574	1.8	96
150	1551	1.7	72	1410	3.5	74	1377	2.3	84
200	1400	2.6	65	1293	3.7	68	1230	4.4	75
300	1183	2.7	55	1105	2.7	58	1080	3.1	66
450	747	7.5	35	680	9.0	36	851	6.5	52

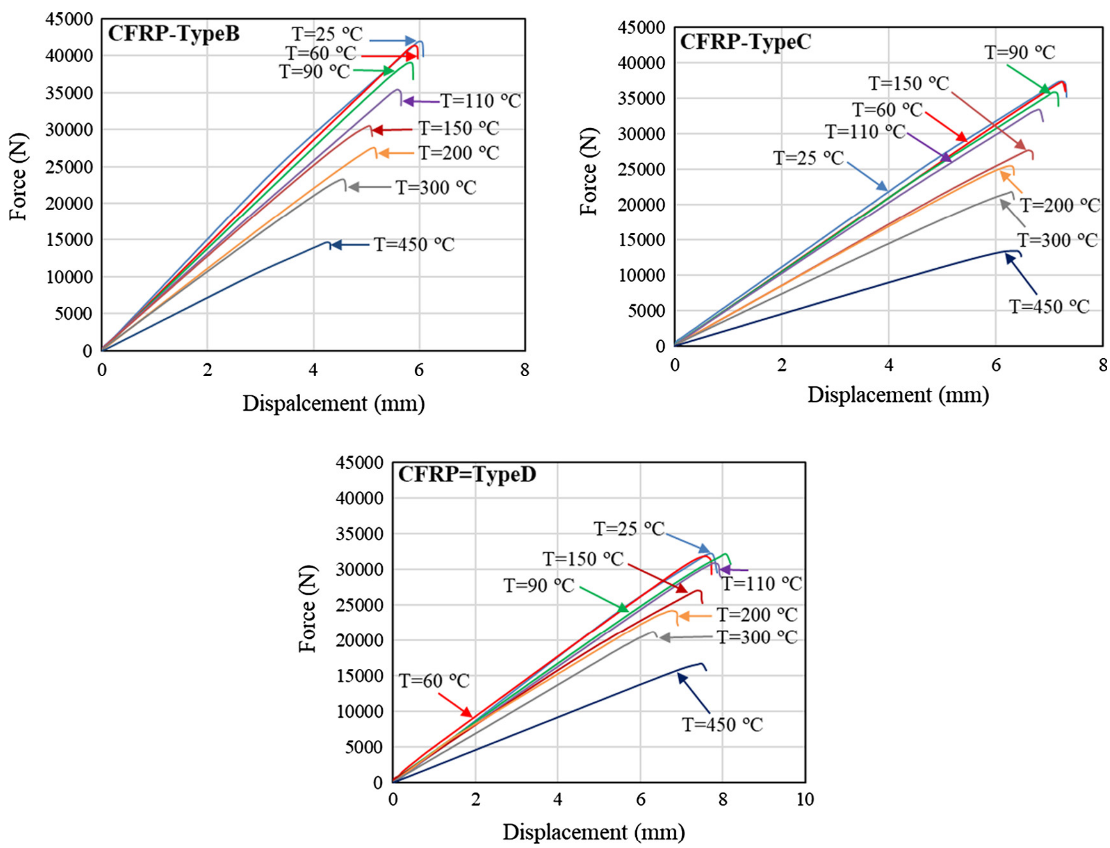

Fig. 7. Selected tensile load versus displacement curves of CFRP bars tested at different temperatures.

The typical tensile load versus the displacement curves of CFRP bars tested at different temperatures are depicted in Fig. 7. Similar to GFRP bars, all CFRP bars showed a linear deformation until they reached approximately their ultimate load, where the bars experienced a slight non-linear deformation followed by a sudden drop due to complete bar failure.

The relationships between the ultimate tensile stress in the CFRP bars and the test temperatures, as well as between tensile stress retention and test temperatures, are shown in Fig. 8. According to the retention-temperature values of the tested CFRP bars, the curves can be separated into three distinct zones. In the first zone, pertaining to the temperatures in the 25–90 °C range, similar to the GFRP bars, no changes in the molecular chain mobility of the matrix were noted. Therefore, the ultimate tensile strength remained almost stable. The second zone (between 90 °C and 150 °C) is also similar to that observed for the GFRP bars, as the test temperatures are close to the T_g values of the resins (T_g = 105 °C for vinylester and T_g = 110 °C for epoxy resin). Hence, a considerable decrease in the ultimate tensile strengths of CFRP bars was observed due to the fiber and matrix separation. In the third zone, pertaining to the temperatures above 150 °C, the ultimate tensile strength continued to exhibit a considerable decline in tensile strength (with a slighter rate compared to the second zone).

The tensile strength results pertaining to the CFRP bars examined in this study are in accordance with those reported by other authors [17,18]. For example, Wang et al. [18] observed that 9.5 mm-GFRP bar tensile strength declined by about 18%, 45%, and 70% at 100 °C, 200 °C, and 400 °C, respectively. Hamad et al. [17] also investigated the mechanical properties of CFRP bars under elevated temperature conditions and recorded a 9%, 27%, 55%, 71%, and 90% decrease in ultimate tensile strength at 125 °C, 250 °C, 325 °C, 375 °C, and 450 °C, respectively.

As can be seen in Fig. 8b, the critical temperatures of CFRP bars were obtained approximately about 330 °C, 360 °C, and 450 °C, for Type B, Type C, and Type D bars, respectively. These results are higher than the critical temperature used for 9.5 mm CFRP bars

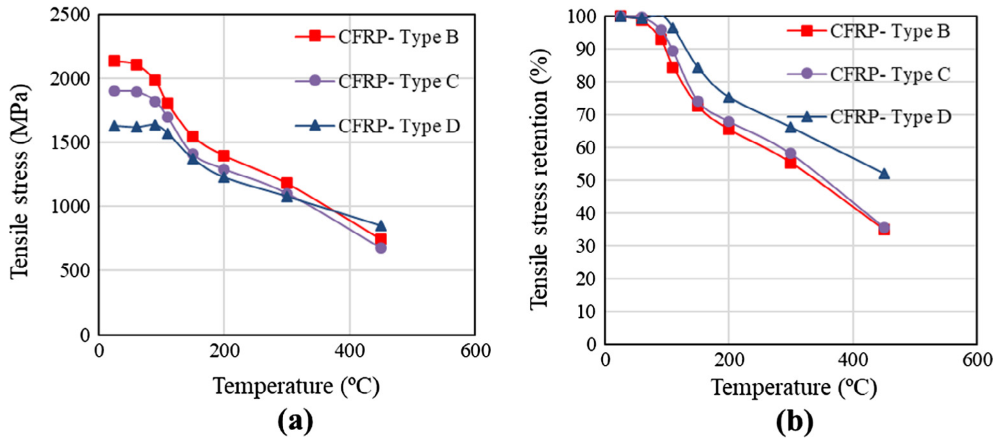

Fig. 8. CFRP bars' test results: a) Ultimate tensile stress vs the test temperatures, and b) Ultimate tensile stress retention vs test temperatures.

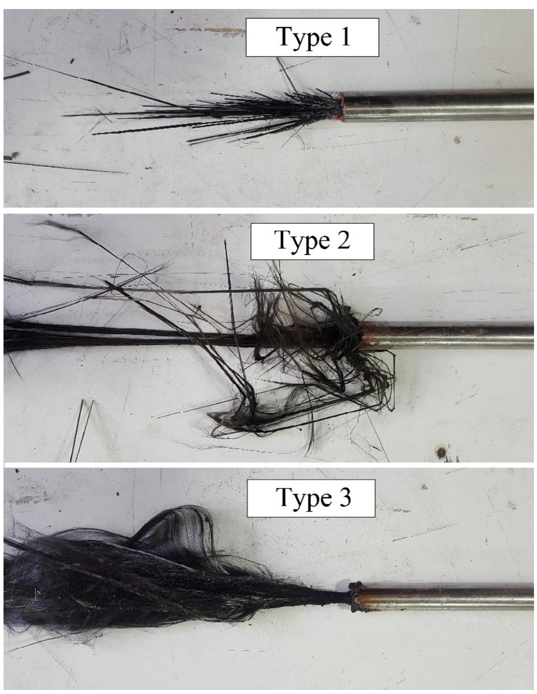

Fig. 9. Failure modes of GFRP bars after tensile tests under elevated temperatures.

in the study of Kodur and Bisby [23] on FRP-RC structures (a critical temperature of 250 °C used in their study).

Fig. 8b reveals that at the same fiber-to-resin ratio, the CFRP bars incorporating vinylester resin (Type D) experienced a lower tensile strength reduction at elevated temperatures compared to CFRP bars in which epoxy resin was used (Type C). This discrepancy may be due to superior vinylester thermal properties, such as decomposition temperature. Furthermore, the similar reduction in tensile strength for Type B and Type C reveals that (if the force transfer between the fibers and resin is provided by the appropriate fiber to matrix ratio) the fiber to resin ratio does not affect the bars' tensile strength at elevated temperatures.

As can be seen in Fig. 9, CFRP bar failure modes were different from those observed in the GFRP bars. However, similar to GFRP bars, CFRP bars experienced three distinct failure modes as a function of test temperature (Fig. 9). The first, second, and third failure types shown in Fig. 9, corresponded to 25–150 °C, 150–300, and >300 °C temperature ranges, respectively.

4. ANOVA (Analysis of Variance)

In order to determine the contribution of different parameters to the ultimate tensile strength of different FRP bars and examine the differences among group means, a two-way analysis of variance (ANOVA) was performed. The ANOVA results of the tensile tests performed on GFRP bars under elevated temperatures, based on the retention values, are presented in Table 4. The SS is the sum of squares of the deviations of all the observations from their mean; The df is the number of degrees of freedom associated with the sample variance; The MS is the mean square, which is obtained by dividing the sum of squares by the respective degrees of freedom; The F is defined as the variation between sample means (Mean Square Between) to the variation within the samples (Mean Squared Error); The P value is the probability that an equal amount of variation in the dependent variable would be observed in the case that the independent variable does not affect the dependent variable; The F crit is an indicator corresponds to the p value (F crit < F shows that the variable has significant effect on the results).

As $p < 0.05$ (F crit < F) was obtained for both temperature and diameter, the contribution of these variables is statistically significant and must be considered when evaluating the tensile strength under fire conditions. On the other hand, the p-value > 0.05 obtained for the interaction of these variables indicates that it has a limited effect on the results. Furthermore, as expected, the

Table 4
Resulting ANOVA table for GFRP bars-tested under temperatures of 25–450 °C.

Source of Variation	SS	df	MS	F	P-value	F crit	Contribution (%)
Temperature	2.979169	6	0.496528	157.0273	3.65E-33	2.265567	82.5
Bar diameter	0.356317	3	0.118772	37.56186	1.95E-13	2.769431	9.8
Interaction	0.097425	18	0.005413	1.711714	0.064583	1.791158	2.6
Error	0.177075	56	0.003162				4.9
Total	3.609986	83					100

Table 5
Resulting ANOVA table for GFRP bars-tested under temperatures of 110–450 °C.

Source of Variation	SS	df	MS	F	P-value	F crit	Contribution (%)
Temperature	1.255081	4	0.31377	73.90704	6.22E-18	2.605975	67
Bar diameter	0.424255	3	0.141418	33.31042	5.82E-11	2.838745	22
Interaction	0.018065	12	0.001505	0.354584	0.971827	2.003459	0.9
Error	0.169819	40	0.004245				9.1
Total	1.86722	59					100

Table 6
Resulting ANOVA table for CFRP bars tested under temperatures of 25–450 °C.

Source of variation	SS	df	MS	F	P-value	F crit	Contribution (%)
Temperature	2.413391	6	0.402232	833.4188	5.36E-42	2.323994	93.61
Bar's properties	0.112205	2	0.056103	116.2438	7.57E-18	3.219942	4.35
Interaction	0.03223	12	0.002686	5.565031	1.39E-05	1.991013	1.25
Error	0.02027	42	0.000483				0.79
Total	2.578097	62					100

Table 7
Resulting ANOVA table for CFRP bars tested under temperatures of 110–450 °C.

Source of Variation	SS	df	MS	F	P-value	F crit	Contribution (%)
Temperature	1.227926	4	0.306981	636.6799	1.55E-28	2.689628	89.03
Bar's properties	0.126895	2	0.063448	131.5904	1.41E-15	3.31583	9.20
Interaction	0.009803	8	0.001225	2.541497	0.030427	2.266163	0.71
Error	0.014465	30	0.000482				1.04
Total	1.379089	44					100

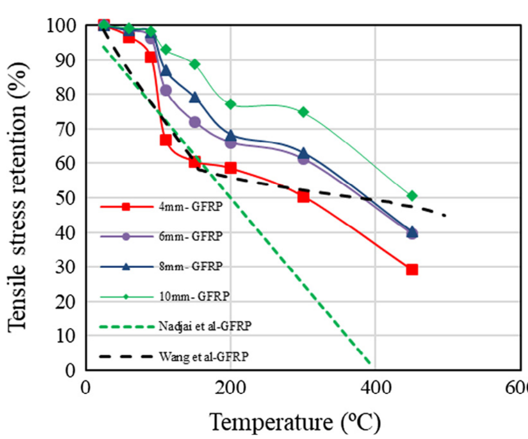

Fig. 10. Comparison between the experimental results and predicted values of GFRP tensile tests at different temperatures.

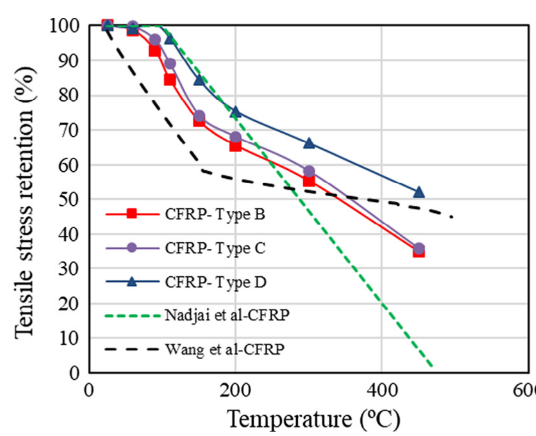

Fig. 11. Comparison between the experimental results and predicted values of CFRP tensile tests at different temperatures.

significant difference between the contributions of temperature and diameter parameters (82% compared to 10%) indicates that the results are minimally affected by the bar diameter. However, the ANOVA analyses performed on the results obtained at temperatures above 110 °C (Table 5) revealed that the effect of bar diameter on the results increases with the increase in the test temperature, whereby the contribution of bar diameter increased from 10% to 22%.

To consider the effects of different parameters on the CFRP tensile test results, namely, temperature, and CFRP bar thermal properties (glass transition temperature T_g, and decomposition temperature T_d), a two-way ANOVA analysis was performed and the results are presented in Table 6. Given that the p-values related to temperature, bar properties, and their interaction are less than 0.05, it is evident that these variables and their interaction must be considered when determining the tensile strength reduction in FRP bars under fire conditions. However, the contribution of temperature is significantly greater compared to that of the bar properties (93% compared to 4%). On the other hand, the ANOVA analysis performed on the results obtained at temperatures exceeding 110 °C (Table 7) revealed that the effect of bar properties on the results increases with test temperature. In fact, the contribution of bar properties increased from 4% to 9%, while that of temperature decreased from 93% to 89%.

5. Comparison between the researchers' predictions and experimental results

In this section, the ultimate tensile strength of FRP bars subjected to elevated temperatures is compared to the predictions yielded by models developed by Nadjai et al.[29] and Wang et al. [30]. Based on the work of Nadjai et al. [29], the residual tensile strength of GFRP and CFRP materials subjected to elevated temperatures can be calculated by applying Eqs. (1) and (2), respectively.

$$\frac{f_{u,t}}{f_u} = \begin{cases} 1 - 0.0025T, & 0 \leqslant T < 400 \\ 0, & T \geqslant 400 \end{cases} \tag{1}$$

$$\frac{f_{u,t}}{f_u} = \begin{cases} 1, & 0 \leqslant T < 100 \\ 1.267 - 0.00267T, & 100 \leqslant T < 475 \\ 0, 475 \leqslant T \end{cases} \tag{2}$$

where $f_{u,t}$ is the tensile strength at temperature T and f_u is the tensile strength at ambient temperature. According to the model presented by Wang et al. [30], the stress–temperature relationship in FPR composites can be determined by Eq. (3):

$$\frac{f_{u,t}}{f_u} = \begin{cases} 1 - \frac{(T-22)^{0.9}}{200}, & 22 \leqslant T < 150 \\ 0.59 - \frac{(T-150)^{0.7}}{490}, & 150 \leqslant T < 420 \\ 0.48 - \frac{(T-420)^{1.8}}{76000}, & 420 \leqslant T < 706 \end{cases} \tag{3}$$

Fig. 10 shows the comparisons between the ultimate tensile strength retention of GFRP bars obtained in this study and those predicted by other researchers [29,30]. As can be seen from the data, both research groups provided very conservative estimates for temperatures below 90 °C. For temperatures between 90 ° and 150 °C, the predictions were more accurate, especially for 4 mm-GFRP bars. However, they were conservative in predicting the ultimate tensile strength retention of 6, 8 and 10 mm-bars. As is apparent, Nadjai et al. [29]'s equation is not appropriate for temperatures exceeding 150 °C, while Wang et al. [30]'s equation is more accurate in this temperature range.

The comparisons between the ultimate tensile strength retention of CFRP bars obtained in this study and those predicted by other researchers are shown in Fig. 11. The equation Nadjai et al. [29] presented for CFRP bars is more accurate than the equation for GFRP bars at temperatures below 300 °C. However, this equation does not predict the results obtained for CFRP bars at temperatures higher than 300 °C properly. On the other hand, as Wang et al. [30] presented the same equation for both GFRP and CFRP bars, similar to GFRP bars, their predictions were conservative at temperatures below 200 °C and more accurate at higher temperatures.

Generally, these results vary because the mechanical and thermal properties of FRP bars can be differ widely cross studies. However, further experimental studies are necessary to revise the current equations so they provide more accurate relations. Additional parameters should be considered as well, such as bar diameter and thermal characteristics of the bars. Therefore, models, which consider additional parameters to the temperature may be more capable of obtaining accurate results.

6. Conclusions

This study is part of an ongoing research program concerning the performance of FRP composites under fire conditions; it seeks to improve the accuracy of the reduction factors and, thus, the design procedure for members reinforced with such materials. This paper focuses on the tensile properties of different types of GFRP and CFRP bars under elevated temperatures. Based on the results obtained, the following conclusions could be drawn:

(1) At temperatures ranging from ambient temperature to 90 °C, which were below the bars' glass transition temperature, the ultimate tensile strength appeared to be almost stable. This showed that temperatures below the FRP bars' T_g slightly affect their tensile behavior.
(2) At temperatures between 90 °C and 150 °C, which were near and above the FRP bars' T_g, the bars' tensile strength decreased significantly due to the change in the molecular bonds and, consequently, fiber/matrix debonding.
(3) At temperatures ranging from 150 °C to 300 °C, the GFRP bars experienced a lower reduction in their tensile strength than the CFRP bars.
(4) At very high temperatures (>300 °C), due to the significant thermal degradation of both types of FRP bars (glass as well as carbon), the ultimate tensile strengths of the bars decreased considerably (about 60–70%).
(5) The critical temperatures of GFRP bars were obtained about 300 °C, 375 °C, 377 °C, and 450 °C for 4, 6, 8, and 10 mm bars, respectively.
(6) The critical temperatures of CFRP bars were obtained approximately about, 330 °C, 360 °C, and 450 °C, for Type B, Type C, and Type D bars, respectively.

The results presented in this study provide a better understanding of the tensile strength reduction of FRP bars under elevated temperatures. However, more detailed experimental data are needed to modify the current equations presented for FRP bars under elevated temperatures.

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